CS480 No-Code Toolkit Documentation Guide

Intro

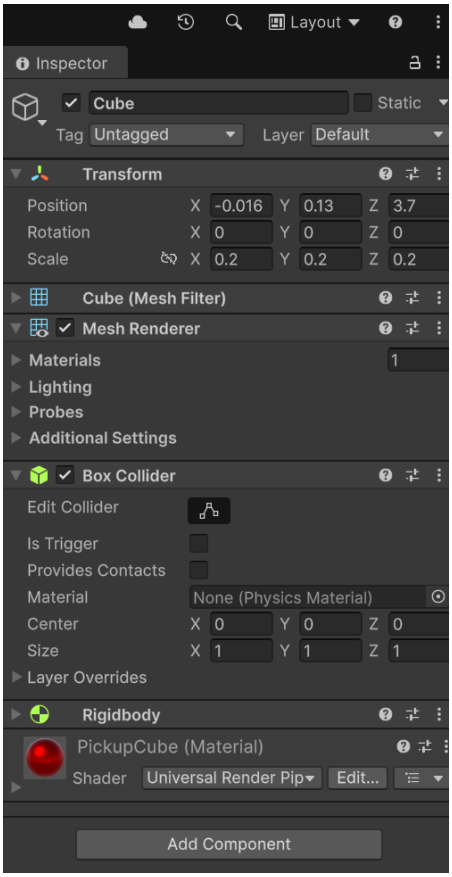
This toolkit is created with the purpose of allowing creators to make interactions within Unity without needing to write a single line of code. By simply adding one of the scripts to an object, you can decide how an object interacts with the player or other objects. The script attached will give you various fields that you can fill out to shape the functionality of your objects, such what to do when a trigger is entered, how long something should wait before acting, and the order that you want your actions to take, just to give a few examples.

Grab Interactable

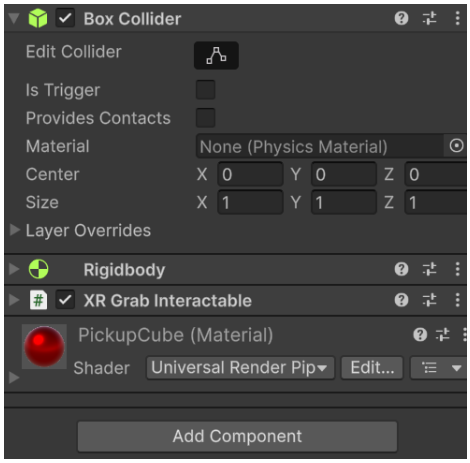
This allows you to pick up or "grab" objects. This is used when you want the player to interact with or move objects around the scene, like picking up an apple on a table.

Setup

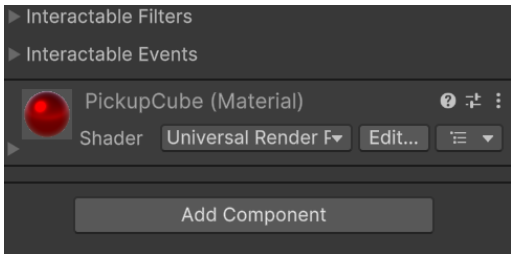
1. Click on the desired object in the **Hierarchy** on the left side of the screen. The look at the **Inspector** on the right side of the screen.
2. Click the **Add Component** button.



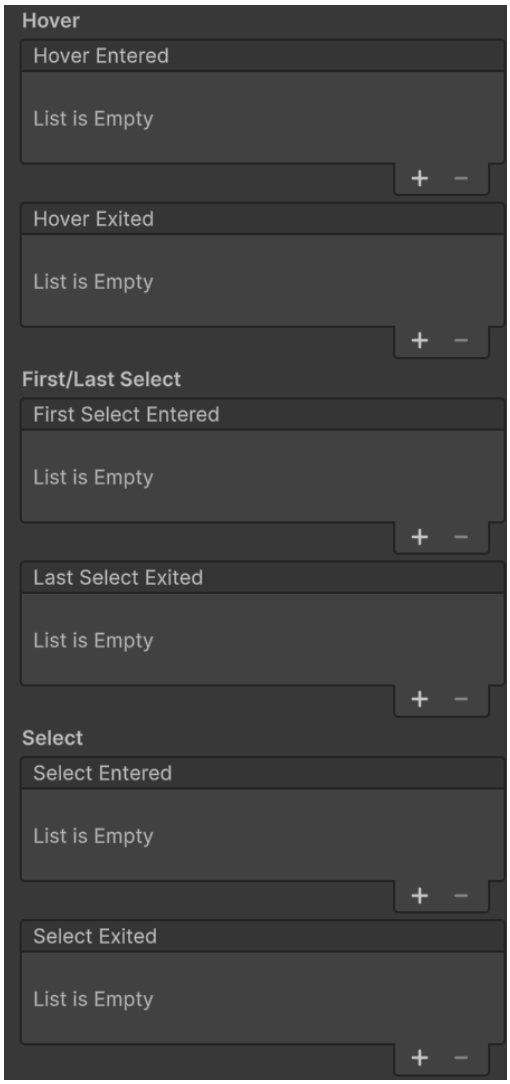
3. In the search bar, type “XR Grab Interactable” and select the script that matches the title. The script should now be one of the components.



4. Open the dropdown menu of the script and look for **Interactable Events**.



5. You should see various windows with different titles. These are the different ways you can add customization to grabbable objects. For example, you can play a sound when the object in question is selected.



Mistakes

There aren't too many gotchas with the grabbable script because it's already built into Unity. I would say if you want to add extra functionality, make sure to go to **Interactable Events** and not **Interactable Filters**.

On Trigger Action

This allows actions to occur when some interacts with an invisible trigger. This can be used when you want something to occur after the player does something, like opening a door after standing on a platform.

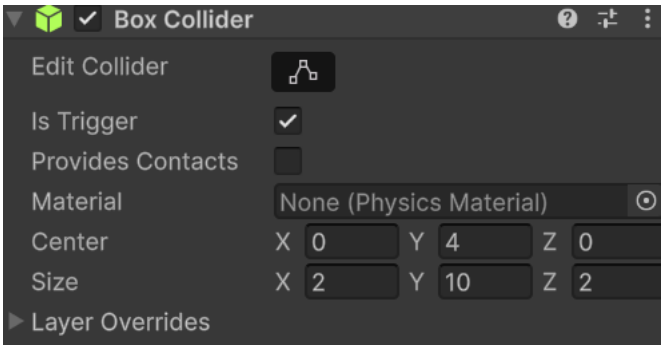
Steps

1. Your object needs to be given a trigger to operate, so in the **Hierarchy**:
   Select your object -> Right-click -> Create Empty
   This will create a new object that should look like it's inside your object. Name it something related to triggers, like **[YOUR SHAPE]Trigger**.

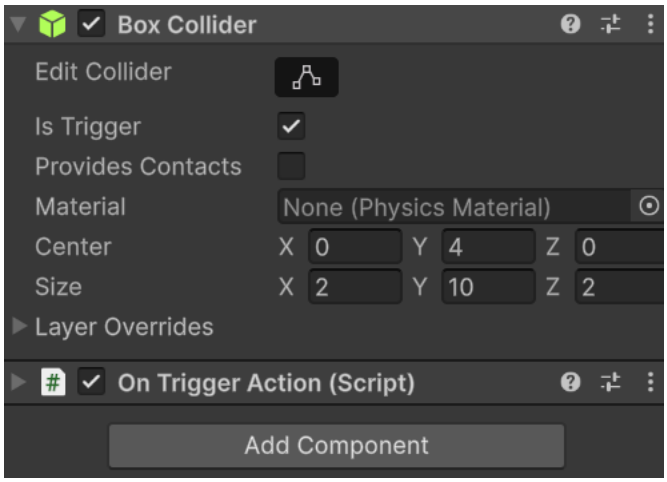


2. In the **Inspector** of the trigger, look inside of the **Box Collider**. You will see two things you want to check:
   - Edit Collider
   - Is Trigger

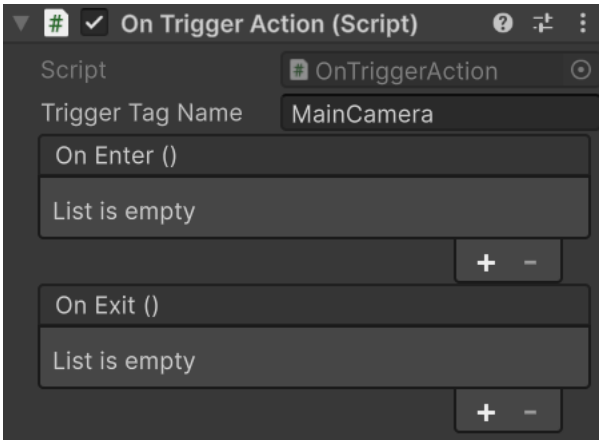
   Edit Collider is going to edit the area that detects the player. In the main window, change the proportions of the box so the player can be detected. This likely means making the box taller so it's eye-level with the player. **Is Trigger** should be checked so the trigger knows it needs to look for something.



3. In the **Inspector**, go to **Add Component**
4. In the search bar, type “On Trigger Action” and select the script that matches the title. The script should now be one of the components.

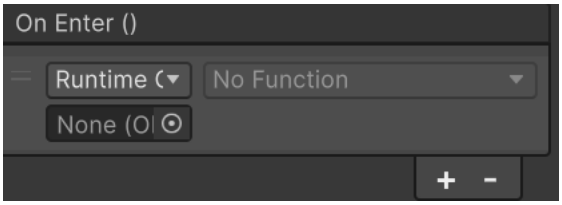


5. Open the dropdown of the script. You should see two tabs:
   - On Enter
   - On Exit

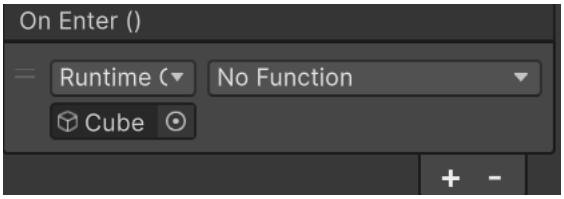


   These are where you will put your actions after either trigger. However, they should both say “List is empty”. On the **On Enter** tab, click the “+” sign at the bottom of it.

6. This should add some boxes that you will use to create the trigger action.



7. Now let's say you want the trigger to make your object disappear when you come into contact with it. To do this, you will need to drag your desired object **(NOT THE TRIGGER)** into object window. It currently says "None", but will change once you put the object in.



8. Next, to make it disappear, we will need to change the function. On the dropdown that says "No Function", click it and do the following:
   GameObject -> SetActive (bool)

Settings

	On Enter	**On Exit**
Object	Anything you want	Anything you want
Function	Keep it simple, like making something appear or disappear, or play a sound	Again, keep it simple

Mistakes

- Make sure you create another object to act as trigger. Using the object itself will not have any effect
- When dragging objects into the script, use the object itself, not the trigger
- If your action doesn't seem to be triggering, check your collider to ensure it's eye-level with the player

Timed Action

This will allow an action to occur after a set period of time, or repeat an action at a regular interval. This could be used to simulate different things like a countdown by ticking every second, or setting a timer for a task that the player must complete within a set timeframe.

Steps

1. Your object needs to be given a trigger to operate, so in the **Hierarchy**:

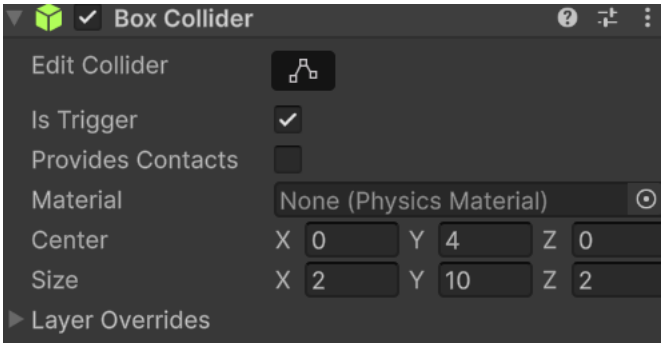
Select your object -> Right-click -> Create Empty

This will create a new object that should look like it's inside your object. Name it something related to triggers, like **[YOUR SHAPE]Trigger**.

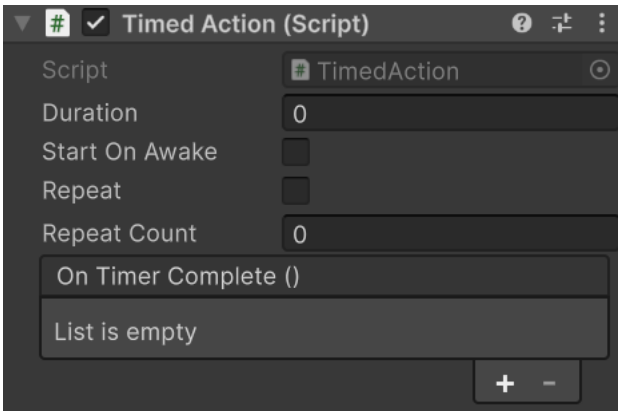


2. In the **Inspector** of the trigger, look inside of the **Box Collider**. You will see two things you want to check:
   - Edit Collider
   - Is Trigger

   Edit Collider is going to edit the area that detects the player. In the main window, change the proportions of the box so the player can be detected. This likely means making the box taller so it's eye-level with the player. **Is Trigger** should be checked so the trigger knows it needs to look for something.



3. In the **Inspector**, go to **Add Component**
4. In the search bar, type "Timed Action" and select the script that matches the title. The script should now be one of the components.
5. Open the dropdown menu and you should see 4 notable fields:
   - Duration
   - Start On Awake
   - Repeat
   - Repeat Count



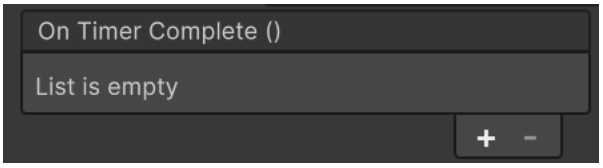
Duration determines how long the timer duration is before the action happens.

Start On Awake means the timer will start as soon as the scene loads.

Repeat determines if the timer will restart after it runs out.

Repeat Count takes the functionality of repeat and does a finite number of times. (If you used the number 3, the timer would repeat only 3 times. The number 0 means it will repeat forever)

6. Right below these fields you will see a tab named “On Timer Complete”. Click the “+” button at the bottom.



7. In this box is where you will insert the object you want to perform an action, and what the action is, using the object field and the function field.

Settings

Duration	Whatever you want, but try to keep it on the smaller side, like 1 to 15
Start On Awake	Depends on if you want the timer to immediately start or if you want to initiate it with something else
Repeat	Depends on if you want the action to occur more than once
Repeat Count	Any number you feel is appropriate for the situation. Remember, 0 means it repeats forever

Mistakes

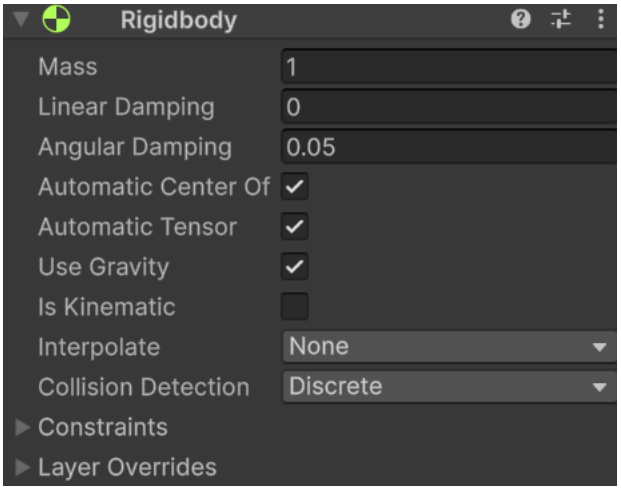
- If you don't want the timer to immediately start, don't forget to turn off **Start On Awake**

Collide Action

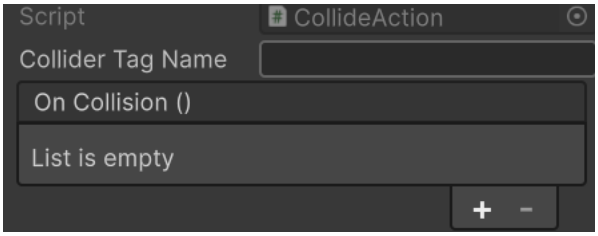
This will allows objects to directly interact with and affect each other. This could be used like making an object disappear when it gets hit by another, or it could play a sound in the same scenario.

Steps

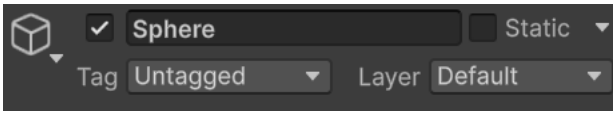
1. Collisions work with the object directly so we don't need to make a trigger. Select your object of choice.
2. In the **Inspector**, you want to ensure that your object has a **RigidBody** component. This is how objects will detect each other. If your object doesn't have one, do the following:
   Add Component => Search "RigidBody" (NOT RigidBody 2D) => Select it
   This will add the RigidBody component to the object.



3. Next, you want to add the **Collide Action** script. Follow these steps:
   Add Component => Search "Collide Action" => Select it
   This should add the script to your object

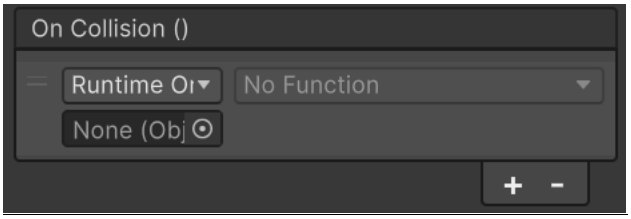


4. The only field here is “Collider Tag Name”. This means that any object with that tag will be detected by your object. You can make and assign tags to various objects at the top of the **Inspector** window.



To make a new tag, click the “Tag” dropdown and do the following:
Add Tag => Click the “+” sign on the “Tags” field => Insert a new name => Save

5. Go back to the script and you will see a tab named “On Collision”. Click the “+” button on the tab. This is where you will input your object and the action to take when it collides with a target.



Settings

Collider Tag Name	This should match the tag of the target object. You can change the target object’s tag or create a new one with step 4 above.

Mistakes

- Colliders use the object, **NOT A TRIGGER**. The object itself is what gets acted on, so the script should go with that.
- At least one of your objects will require a RigidBody. That’s what will detect the target object(s).
- For any objects you want to interact, if the target object tag doesn’t match what your object is searching for, nothing will happen.

Sequence Action

This will allow you to string actions together and space them out with timers. This could be used to simulate something like unlocking a door with a key, or playing a chord of music by introducing each note.

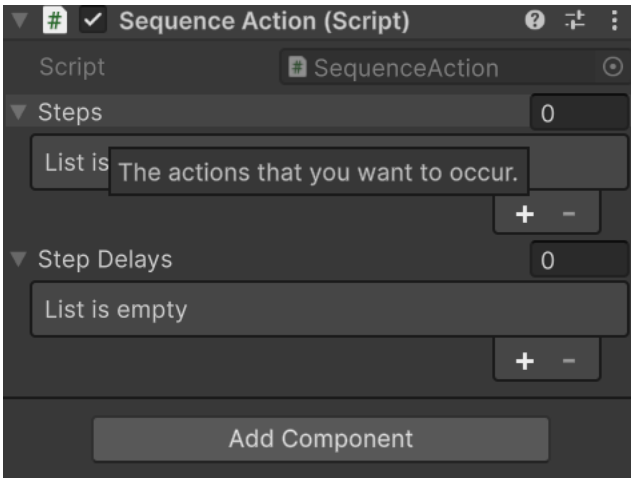
Steps

1. Create a new object that can act as the host for the script. It can be empty or a real object.
2. In the object, do the following:
   Add Component => Search “Sequence Action” => Select it
   The script should now be part of the object

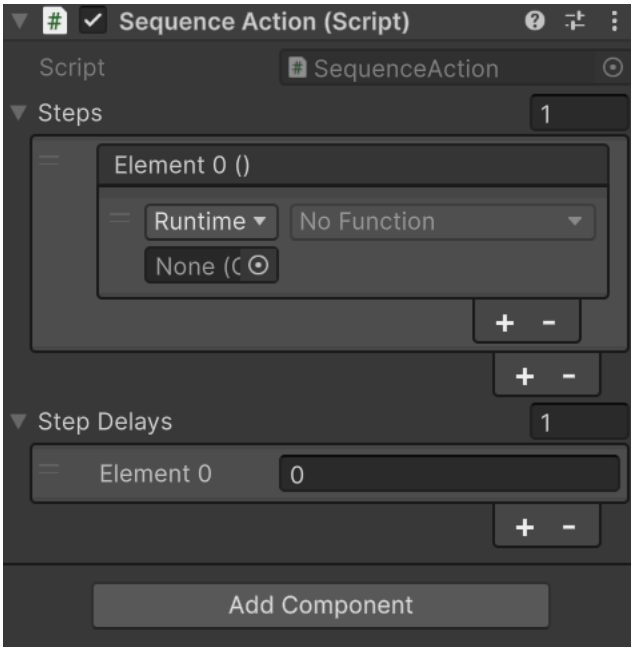


3. If you open the script, you will see to dropdowns:
   - Steps
   - Step Delays

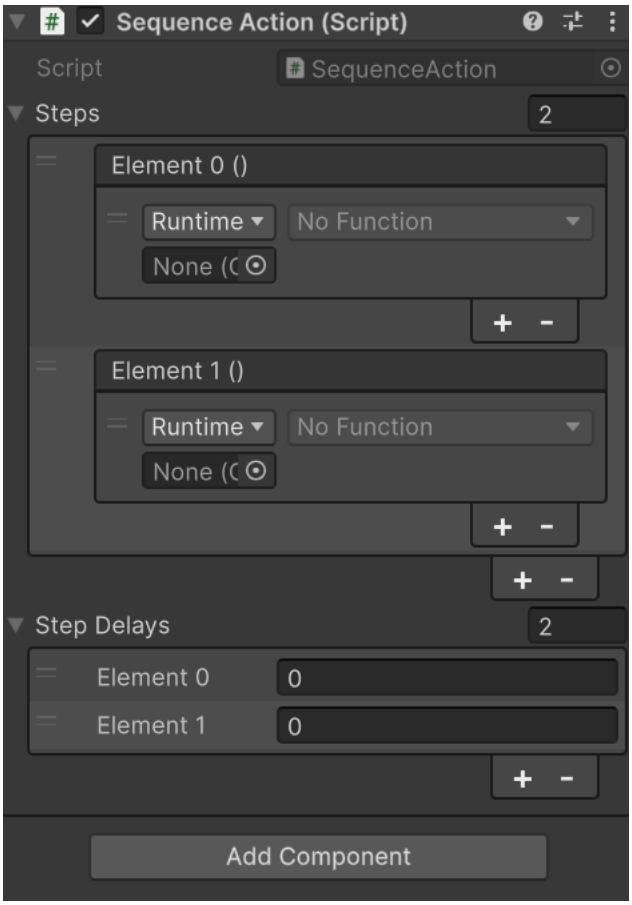
   These are where you will put the actions you want to occur and the amount of time before the action occurs respectively.



4. Click the “+” button on both of the dropdowns. You will see that **Step Delays** has a field with 0 in it. In **Steps**, you will see “Element 0” and “List is empty”. Hit the “+” button again, but this time on the element (inside) instead of the **Steps** box (outside). You should now see that you can alter the things inside the element.



5. This is the setup for one item. Now if you wanted to add another item, you would simply repeat step 4. The only difference is that the second element should already have editable fields when it’s created. This can be done as many times as you want.



6. For the steps, you will drag the desired element into the object box, and select the action to take in the function box, for all of your elements. For the step delays, your number will determine how long to wait before doing the action. So if my first number is 4 and my second is 1, the first action will occur after 4 seconds and the second action will occur after 1 second. If the number is 0, then the action will occur automatically.

Settings

Steps	You will insert the desired object and the action it should do
Step Delays	This is the timer before the action happens

Mistakes

- Clicking the "+" on the element itself will add another function to that element. If that is intentional, disregard this, otherwise ensure you click the "+" on the overall **Steps** tab.

- The number 0 means there is no wait before the action occurs. If you want a delay, make sure to change your number to something greater than 1.

Putting It All Together

In my scene, I have various things with shapes occurring. You can pick up the small squares and throw them around. However, if you get to close to the red one, its trigger makes the black one disappear. You can also pick up the ball and roll it into the cylinder, which is constantly clicking. If the ball hits the cylinder then the cylinder disappears. Finally, I have lovely C major chord that plays after a few seconds, symbolized by three capsules.

